Machine Learning Notes

Day 8 RAG practice document

1. Supervised Learning

Supervised learning uses labelled examples. Classification predicts a category, while regression predicts a numerical value. Model performance should be evaluated on data not used for training.

2. Overfitting

A model overfits when it learns noise or details specific to its training data. It may achieve excellent training performance but perform poorly on new examples. Regularization, more data, and simpler models can help.

3. Train and Test Split

A dataset is commonly divided into training, validation, and test sets. The training set fits model parameters, the validation set supports model selection, and the test set estimates final generalization performance.

4. Embeddings

An embedding maps an item such as a sentence into a vector of numbers. Embeddings can represent semantic relationships, allowing systems to retrieve text that is similar in meaning even when it uses different words.

5. Evaluation

Accuracy is the proportion of correct predictions. Precision measures how many predicted positives are correct, while recall measures how many actual positives were found. The appropriate metric depends on the cost of different errors.